

D869

Biochemistry I

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A sports-drink company's research team, which just moved into a new facility, studies how two amino acids join to form a dipeptide in its protein-blend product. Besides the dipeptide, what does the reaction produce?

- A. Only a hydrogen bond, with no by-product
- B. One molecule of water
- C. Nothing; one molecule of water is used up
- D. A phosphodiester bond and water

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

A nutrition lab runs its tests at 37 °C to match body temperature, and each sample was weighed to 0.1 mg. It compares glucose, cholesterol, a triglyceride, and a long-chain fatty acid. Which dissolves best in water?

- A. The long-chain fatty acid, because it has a polar head
- B. Cholesterol, because it has an -OH group
- C. Glucose
- D. The triglyceride

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3. [TRANSFER — reversed stem / different variable]

A molecule dissolves readily in hexane but barely in water. Which structural feature best explains this?

- A. A long hydrocarbon chain with few polar groups
- B. A net electrical charge
- C. A high molar mass
- D. Many hydroxyl groups

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

An enzyme purified from bovine liver and stored at -80°C is treated with a reducing agent such as β -mercaptoethanol. Which bonds in the enzyme does this treatment break?

- A. Hydrogen bonds only
- B. Peptide bonds in the backbone
- C. Disulfide bridges between cysteine side chains
- D. All bonds, breaking the enzyme into free amino acids

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

An enzyme bought from a supplier in 5-mg vials is assayed at pH 7.4. Its V_{max} is $120\ \mu\text{mol}/\text{min}$, and at a substrate concentration of $2.5\ \text{mM}$ the rate is $60\ \mu\text{mol}/\text{min}$. What is K_{m} ?

- A. $2.5\ \text{mM}$
- B. $120\ \mu\text{mol}/\text{min}$
- C. $5.0\ \text{mM}$
- D. $60\ \mu\text{mol}/\text{min}$

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6. [TRANSFER — reversed stem / different variable]

An enzyme has $K_{\text{m}} = 4.0\ \text{mM}$ and $V_{\text{max}} = 100\ \mu\text{mol}/\text{min}$. What is the rate at $[S] = 12\ \text{mM}$?

- A. $100\ \mu\text{mol}/\text{min}$, because $[S]$ is above K_{m}
- B. $50\ \mu\text{mol}/\text{min}$
- C. $300\ \mu\text{mol}/\text{min}$
- D. $75\ \mu\text{mol}/\text{min}$

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A muscle cell at rest has high ATP and low AMP levels. The cell also contains plenty of stored glycogen. What happens to the rate of glycolysis?

- A. It is unchanged, because enzymes respond only to substrate concentration.
- B. It slows, because ATP allosterically inhibits PFK-1.
- C. It speeds up, because ATP is a substrate of PFK-1.
- D. It slows, because glucose has run out.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A yeast culture in a sealed 5-liter bioreactor held at 30 °C breaks down glucose through the ten enzyme-catalyzed steps of glycolysis. What is the net ATP yield of glycolysis per glucose?

- A. About 30–32 ATP
- B. 4 ATP
- C. 10 ATP, one per step
- D. 2 ATP

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9. [TRANSFER — reversed stem / different variable]

A cell breaks down 5 glucose molecules by glycolysis only. What are the net ATP and pyruvate totals?

- A. 20 ATP and 10 pyruvate
- B. 10 ATP and 10 pyruvate
- C. 10 ATP and 5 pyruvate
- D. 5 ATP and 5 pyruvate

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

In a liver cell's mitochondrial matrix, one glucose molecule has been fully converted to acetyl-CoA, which then enters the citric acid cycle. The cell is well supplied with oxygen. How many CO₂ molecules does the citric acid cycle itself release from that glucose?

- A. 2
- B. 3
- C. 4
- D. 6

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A rodenticide sold in 1 kg tubs contains a compound that blocks complex IV of the electron transport chain. Its label warns it is fast-acting. In an exposed cell, what is the direct effect?

- A. Oxygen use falls and ATP synthase makes much less ATP.
- B. Glycolysis stops at once, because all ATP depends on oxygen.
- C. Only the citric acid cycle is affected, because it runs in the matrix.
- D. Oxygen use rises as the cell tries to compensate.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12. [TRANSFER — reversed stem / different variable]

An uncoupler makes the inner mitochondrial membrane leak protons. What happens to oxygen use and ATP production?

- A.** The electron transport chain stops completely.
- B.** Oxygen use rises and ATP production rises.
- C.** Oxygen use falls and ATP production falls.
- D.** Oxygen use rises and ATP production falls.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	—	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer B Concept 1

A sports-drink company's research team, ~~which just moved into a new facility~~, studies how two amino acids join to form a dipeptide in its ~~protein blend product~~. Besides the dipeptide, what does the reaction produce?

Target word: by-product of forming a peptide bond **Decides:** Forming a peptide bond is a condensation reaction.

A	Adjacent category	A peptide bond is covalent; hydrogen bonds are weak attractions.
B	Correct	Condensation removes -OH from one amino acid and -H from the other.
C	Inverted operation	That describes hydrolysis, which breaks the bond.
D	Confusion set	Phosphodiester bonds join nucleotides, not amino acids.

2. Answer C Concept 2

A nutrition lab ~~runs its tests at 37 °C to match body temperature~~, and ~~each sample was weighed to 0.1 mg~~. It compares glucose, cholesterol, a triglyceride, and a long-chain fatty acid. Which dissolves best in water?

Target word: most water-soluble molecule **Decides:** Glucose has five -OH groups and no long nonpolar chain.

A	Adjacent category	Its long nonpolar tail dominates; it forms micelles rather than dissolving freely.
B	Confusion set	One -OH on a large nonpolar ring system does not make it water-soluble.
C	Correct	Many hydroxyl groups hydrogen-bond with water.
D	Inverted operation	Triglycerides are the least polar of the four.

3. Answer A Concept 2 · transfer

A molecule dissolves readily in hexane but barely in water. Which structural feature best explains this?

Target word: reason for nonpolar solubility **Decides:** Like dissolves like.

A	Correct	Nonpolar chains interact well with hexane and poorly with water.
B	Confusion set	Charged groups favor water, not hexane.
C	Adjacent category	Size alone does not decide polarity; large sugars dissolve in water.
D	Inverted operation	Hydroxyl groups make a molecule more water-soluble.

4. Answer C Concept 3

An enzyme ~~purified from bovine liver and stored at -80 °C~~ is treated with a reducing agent such as β -mercaptoethanol. Which bonds in the enzyme does this treatment break?

Target word: bond broken by a reducing agent **Decides:** Reducing agents break S-S bridges between cysteines.

A	Confusion set	Heat or urea disrupt hydrogen bonds; a reducing agent targets disulfides.
B	Adjacent category	Peptide bonds are primary structure and need hydrolysis to break.
C	Correct	Reduction converts S-S back to two -SH groups.
D	Absolute language	Denaturing leaves the amino acid sequence intact.

5. Answer A Concept 4

An enzyme bought from a supplier in 5-mg vials is assayed at pH 7.4. Its V_{max} is 120 $\mu\text{mol/min}$, and at a substrate concentration of 2.5 mM the rate is 60 $\mu\text{mol/min}$. What is K_m ?

Target word: K_m **Decides:** 60 is half of 120, so $[S]$ at that rate equals K_m .

A	Correct	K_m is the $[S]$ giving half of V_{max} .
B	Confusion set	That is V_{max} .
C	Inverted operation	Doubles $[S]$ as if K_m were where v reaches V_{max} .
D	Wrong stem number	That is the rate, not a concentration.

6. Answer D Concept 4 • transfer

An enzyme has $K_m = 4.0 \text{ mM}$ and $V_{max} = 100 \mu\text{mol/min}$. What is the rate at $[S] = 12 \text{ mM}$?

Target word: solve for v **Decides:** $v = V_{max}[S] / (K_m + [S])$.

A	Absolute language	v only approaches V_{max} at very high $[S]$.
B	Confusion set	That is the rate at $[S] = K_m$, not at 12 mM.
C	Inverted operation	$100 \times 12 / 4$ drops $[S]$ from the denominator and exceeds V_{max} .
D	Correct	$100 \times 12 / (4 + 12) = 1,200 / 16 = 75$.

7. Answer B Concept 5

A muscle cell at rest has high ATP and low AMP levels. ~~The cell also contains plenty of stored glycogen.~~ What happens to the rate of glycolysis?

Target word: effect of energy charge on glycolysis **Decides:** High ATP binds PFK-1's allosteric site and inhibits it.

A	Absolute language	Allosteric regulators change enzyme activity.
B	Correct	Feedback inhibition matches production to demand.
C	Inverted operation	ATP is a substrate, but at high levels it binds the inhibitory site.
D	Camouflage	Glycogen stores are plentiful; the brake is ATP, not lack of fuel.

8. Answer D Concept 6

A yeast culture ~~in a sealed 5-liter bioreactor held at 30 °C~~ breaks down glucose through ~~the ten enzyme-catalyzed steps of~~ glycolysis. What is the net ATP yield of glycolysis per glucose?

Target word: net ATP from glycolysis **Decides:** 4 ATP made minus 2 invested.

A	Adjacent category	That is full aerobic respiration, not glycolysis alone.
B	Dropped element	Gross yield; leaves out the 2 ATP invested.
C	Camouflage	The number of steps does not decide the yield.
D	Correct	$\text{Net} = 4 - 2 = 2 \text{ ATP per glucose}$.

9. Answer B Concept 6 • transfer

A cell breaks down 5 glucose molecules by glycolysis only. What are the net ATP and pyruvate totals?

Target word: scale up the per-glucose yield **Decides:** Per glucose: net 2 ATP and 2 pyruvate.

A	Dropped element	Uses gross 4 ATP per glucose.
B	Correct	$5 \times 2 = 10$ of each.
C	Confusion set	Each glucose splits into two pyruvate.
D	Wrong stem number	Echoes the glucose count.

10. Answer C Concept 7

~~In a liver cell's mitochondrial matrix, one glucose molecule has been fully converted to acetyl-CoA, which then enters the citric acid cycle. The cell is well supplied with oxygen.~~ How many CO₂ molecules does the citric acid cycle itself release from that glucose?

Target word: CO₂ from the cycle per glucose **Decides:** Two acetyl-CoA → two turns × 2 CO₂.

A	Dropped element	Counts one turn; glucose gives two acetyl-CoA.
B	Confusion set	That is NADH per turn, not CO ₂ .
C	Correct	$2 \text{ turns} \times 2 \text{ CO}_2 = 4$.
D	Adjacent category	Adds the 2 CO ₂ from pyruvate oxidation, which comes before the cycle.

11. Answer A Concept 8

A rodenticide ~~sold in 1-kg tubs~~ contains a compound that blocks complex IV of the electron transport chain. ~~Its label warns it is fast acting.~~ In an exposed cell, what is the direct effect?

Target word: effect of blocking complex IV **Decides:** Electrons cannot reach O₂, so the chain stops.

A	Correct	No electron flow means no proton pumping and no gradient to drive ATP synthase.
B	Absolute language	Glycolysis is anaerobic and continues.
C	Adjacent category	The block is in the chain; the cycle slows only as NAD ⁺ runs short.
D	Inverted operation	O ₂ is only consumed at complex IV, which is blocked.

12. Answer D Concept 8 • transfer

An uncoupler makes the inner mitochondrial membrane leak protons. What happens to oxygen use and ATP production?

Target word: effect of an uncoupler **Decides:** The chain runs freely but the gradient bypasses ATP synthase.

A	Adjacent category	Uncouplers remove the brake on the chain rather than blocking it.
B	Inverted operation	Leaked protons do not pass through ATP synthase.
C	Confusion set	That is the pattern for a chain inhibitor, not an uncoupler.
D	Correct	Electron flow speeds up; energy is lost as heat instead of ATP.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owled.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	3	
Wrong stem number	A figure copied from the prompt that is not the answer.	2	
Adjacent category	The right idea in the wrong bucket.	8	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	8	
Absolute language	Always, never, guarantee — where judgment applies.	4	
Confusion set	Every option is real; the test is exact role-matching.	9	
Camouflage	Answers a stem fact that was never the question.	2	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Condensation vs. hydrolysis	Building a polymer bond releases water (condensation); breaking it uses water (hydrolysis).	Thinking bond-forming reactions consume water.	Monomers being joined or a polymer being digested.
2	Polarity predicts solubility	Like dissolves like: molecules rich in -OH or charged groups dissolve in water; long hydrocarbon chains dissolve in nonpolar solvents.	Judging solubility by size or by a single polar group on a large nonpolar molecule.	Compare how well several biomolecules dissolve in water.
3	Forces holding tertiary structure	Tertiary structure is held by hydrogen bonds, ionic bonds, hydrophobic interactions, and covalent disulfide bridges; only disulfides are covalent.	Assuming denaturing agents break peptide bonds.	A reducing agent, heat, or a pH change applied to a protein.
4	K_m and V_{max}	K _m is the substrate concentration at half of V _{max} ; $v = V_{max}[S] / (K_m + [S])$.	Reading K _m as a rate, or thinking v reaches V _{max} as soon as [S] passes K _m .	A rate that is exactly half the maximum.
5	Allosteric feedback regulation	End products or energy signals bind allosteric sites to slow or speed key enzymes; high ATP inhibits PFK-1 and high AMP activates it.	Thinking more of a substrate-like molecule always speeds the pathway.	The cell's energy charge is high or low.
6	Glycolysis net yield	One glucose → 2 pyruvate, a net 2 ATP (4 made – 2 invested), and 2 NADH.	Reporting the gross 4 ATP or the whole-respiration total.	"Net ATP from glycolysis."
7	Citric acid cycle output	Each turn oxidizes one acetyl-CoA, releasing 2 CO ₂ and making 3 NADH, 1 FADH ₂ , and 1 ATP/GTP; one glucose gives two turns.	Counting per turn when the question asks per glucose, or adding pyruvate oxidation.	"Per glucose" vs. "per acetyl-CoA."
8	Electron transport and O₂	O ₂ is the final electron acceptor; blocking the chain stops O ₂ use and proton pumping, cutting ATP synthase output, while an uncoupler speeds O ₂ use but cuts ATP.	Confusing an inhibitor with an uncoupler.	A poison that blocks a complex or makes the membrane leak protons.

Course scope (public description): This course delves into the critical role of macromolecules and chemical reactions in living organisms, with a focus on molecular interactions, metabolic pathways, and the regulatory mechanisms that underpin biological processes. The student will start by analyzing macromolecules in organisms and study information on the structure, function, and interactions of key biological macromolecules including carbohydrates, lipids, proteins, and nucleic acids, emphasizing their roles in cellular functions and intercellular interactions. The course then builds to analyzing macromolecules in organisms by building on this foundation, examining the interactions between these macromolecules and the forces driving these interactions, such as intermolecular forces and solubility principles. Finally, the student will explore chemical reactions in organisms through enzyme function and inhibition, metabolic pathways, like glycolysis and the citric acid cycle, and the electron transport chain, as well as the regulation of these pathways and the impact of metabolic disorders. Throughout the course, students will engage in activities and analyses that deepen their understanding of these biochemical processes and their relevance to the functioning of living organisms.